

Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (MICROBIOLOGY)

SEMESTER III EXAMINATION (BACKLOG)

MI 801: GENETIC ENGINEERING AND BIOINFORMATICS

Date: 21.04. 2011

Time: 10:00 AM to 1:00 PM

Total Marks: 70

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A

Total marks: 20

Q1.a. Choose the correct option and put $\sqrt$ mark in front of it:

1. Which one of the following nomenclature is correct for a restriction enzyme?
a) Bamh 1 b) BamH 1 c) BamH I d) *BamH I*
2. Insertional inactivation of reporter gene provides an effective means of recombinant identification. [True] [False]
3. When a foreign DNA (2 Kb) is inserted into the MCS located in λcI gene of λ based vector, then the resulting plaques obtained will be
a) Blue plaques b) White plaques c) Clear plaques d) Turbid plaques
4. W. Arber, H. Smith and D. Nathans in 1978 shared the Nobel Prize for the discovery of
a) Restriction enzymes b) Vector c) Plasmid d) α -complementation
5. mRNA template in the RNA-DNA duplex can be removed by the action of following enzyme
(a) RNAase A (b) RNAase H (c) DNAase (d) endonuclease
6. "RACE" is a method used for the determination of
(a) restriction sites in a gene (b) transcription start of a gene
(c) histone binding sites on DNA (d) none of the above
7. For construction of gene deletions in bacteria the following gene can be used as counter selectable marker in pop-in, pop-out method.
(a) *lacZ* (b) *xylE* (c) *c I* (d) *sac B*
8. Comet assay can be used to identify the following elements in a genome
(a) DNA hypersensitive sites (b) transcription start site
(c) promoters (d) introns
9. Which of the following tool is used to access the biodiversity of the ecological samples
a) Genomics b) Proteomics c) Transcriptomics d) Metagenomics
10. Which of the following PCR technique is used for the analysis of gene expression
a) RT-PCR b) qPCR c) Recombinant PCR d) PCR

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11. In proteomics, antibodies are used for the following application
a) DNA binding analysis b) Single protein detection
c) Multiple protein detection d) Proteome analysis
12. Transcription of transgene can be analyzed by
a) Western Blotting b) ELISA c) Northern Blotting d) X-Ray Crystallography
13. Following is not a Boolean operator
a) AND b) IF c) NOT d) OR
14. RefSeq is a
a) Curated Secondary Database b) Uncurated Secondary Database
c) Primary Database d) Specialized Database
15. TPA stands for
a) Third Party Association b) Third Party Allotment
c) Third Party Activation d) Third Party Annotation
16. NCBI is located at
a) Chicago, USA b) New York, USA
c) Texas, USA d) Bethesda, USA
17. VADAR is a
a) Structure evaluation tool b) Sequence alignment tool
c) ORF detection tool d) RNA-Protein translation tool
18. A good protein structure should have a low R-factor [True] [False]
19. SCOP
a) Compares and classify protein structures b) Visualize protein structures
c) Determines protein Structure d) None of the above
20. Kimura Model corrects
a) Homozygosity b) Homoplasmy c) Heteroplasmy d) Heterozygosity

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PART B

Total marks: 50

Section I

- Q.1.A Describe **any two** of the following. (06)
i) The salient features of plasmid vectors
ii) Phage vectors
iii) Blue-white screening
- Q.1. B Describe in detail about yeast artificial chromosome (YAC). (03)
- OR**
- Q.1. B Draw a schematic diagram involving all the steps in cloning of a 3.5 kb 'chromate reductase' gene in pUC19 vector. (03)
- Q.1. C Define a 'stuffer' fragment. (01)
- Q.2.A Discuss Site directed mutagenesis (04)
- OR**
- Q.2.A Discuss
i) DNAase Hypersensitivity assay
ii) Complementation
- Q.2.B Write a note on **any one** of the following: (03)
i) Immunological screening of expression libraries
ii) Construction of Genomic library
- Q.2.C Explain briefly **any one** the following: (03)
i) Electromobility shift assay
ii) Hydroxyl radical foot printing
iii) Colony hybridization

Section –II

- Q.3. A Write notes on (**Any two**) (06)
i) Functional Proteomics
ii) Site directed mutagenesis
iii) Biomarkers
- Q.3.B Answer the following (**Any two**) (04)
i) Explain the use of Methylated DNA immunoprecipitation as an diagnostic tool.
ii) Give brief account of Burkitt's Lymphoma
iii) Explain comparative genomics.
- Q.4.A What is a database? Describe in brief various types of biological databases (04)
- OR**
- Q.4.A Write a note on the importance of developing various types of tools in bioinformatics (04)
- Q.4.B What is Genbank flat file. (03)
- OR**
- Q.4.C Write a short note on OMIM. (03)
What is TPA in bioinformatics? (01)
- Q.4.D Describe in brief UniProtKB flat file. (02)
- Q.5.A List any six characteristics required to generate acceptable protein structure. (02)
- Q.5.B What is the significance of studying Phylogenetics (03)
- OR**
- Q.5.B Mention various steps involved in Phylogenetic Tree Construction (03)
- Q.5.C What characteristics a good protein structures should have to obtain high quality 3D protein image? (03)
- OR**
- Q.5.C What is the importance of Phylogenetics at the molecular level (03)
- Q.5.D Describe in brief the two Protein structure Classification Schemes (02)